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Physics: Thermodynamics: First Law, Heat Engines & Carnot Cycle

Comprehensive revision notes: Quick, Topper, Chatpata & Practice Questions

Class 12 (Science) • Subject: Physics • Chapter: Thermodynamics: First Law, Heat Engines & Carnot Cycle • Duration: 38 mins lecture

Quick Revision & High-Yield Bullets

Class 12 Physics: Thermodynamics, First Law, Heat Engines & Carnot Cycle Revision Notes

1. First Law of Thermodynamics (Energy Conservation): Heat supplied to a system equals the increase in internal energy plus work done by the system: Q = Delta U + W (differential form: dQ = dU + P dV).
2. Physics Sign Conventions: Heat absorbed by system Q > 0; Heat released Q < 0. Work done BY system (expansion) W > 0; Work done ON system (compression) W < 0.
3. State vs Path Functions: Internal energy U is a state function (U depends solely on absolute temperature T in an ideal gas). In any cyclic process, Delta U_cycle = 0. Heat Q and Work W are path functions.
4. Isothermal Process (Constant T, Delta T = 0): Delta U = 0, so Q = W. Work done: W = nRT ln(V2 / V1) = 2.303 nRT log10(V2 / V1).
5. Adiabatic Process (Zero Heat Exchange, Q = 0): W = -Delta U. Governing relation: P V^gamma = constant. Work done: W = (P1 V1 - P2 V2) / (gamma - 1) = nR (T1 - T2) / (gamma - 1).
6. Isobaric Process (Constant P): Work done W = P(V2 - V1) = nR(T2 - T1). Heat added Q = n Cp Delta T.
7. Isochoric Process (Constant V, dV = 0): Work done W = 0. Heat added Q = Delta U = n Cv Delta T.
8. Heat Engine Efficiency: A cyclic device extracting heat Qh from source at Th, performing work W = Qh - Qc, and rejecting Qc to sink at Tc. Thermal efficiency: eta = W / Qh = 1 - (Qc / Qh).
9. Carnot Engine & Maximum Efficiency: Four-stage ideal reversible cycle (2 isothermals + 2 adiabatics). Efficiency depends solely on reservoir temperatures: eta_carnot = 1 - (Tc / Th). (Temperatures strictly in Kelvin).
10. Mayer’s Relation & Specific Heats: Cp - Cv = R; gamma = Cp / Cv. For monoatomic gas gamma = 5/3 ~= 1.67; for diatomic gas gamma = 7/5 = 1.40.

Key Formulas & Governing Equations Table

Formula Name	Mathematical Expression	Units & Applicability Notes
First Law of Thermodynamics	`Q = Delta U + W = Delta U + Integral  P dV`	Conservation of energy; W > 0 for expansion against surroundings, Q > 0 for heat added. Units: Joules (J).
Internal Energy of Ideal Gas	`Delta U = n Cv Delta T = (f/2) n R Delta T`	f = degrees of freedom (f=3 for monoatomic, f=5 for rigid diatomic). Depends strictly on temperature T.
Isothermal Reversible Work	`W = n R T ln(V2 / V1) = 2.303 n R T log10(V2 / V1)`	Valid for ideal gas at constant temperature T; if V2 > V1 (expansion), W > 0.
Adiabatic Process Governing Law	`P V^gamma = const | T V^(gamma-1) = const | T^gamma P^(1-gamma) = const`	gamma = Cp / Cv > 1. Steeper P-V slope than isothermal curve by factor of gamma.
Adiabatic Reversible Work	`W = (P1 V1 - P2 V2) / (gamma - 1) = n R (T1 - T2) / (gamma - 1)`	Expansion (V2 > V1) causes gas cooling (T2 < T1), with positive work delivered at expense of internal energy.
Heat Engine Thermal Efficiency	`eta = W / Qh = (Qh - Qc) / Qh = 1 - (Qc / Qh)`	Dimensionless ratio (0 <= eta < 1). Qh = heat absorbed from hot reservoir; Qc = heat discharged to cold reservoir.
Carnot Engine Efficiency	`eta_carnot = 1 - (Tc / Th)`	Maximum possible efficiency between reservoirs at Th

Formula Name	Mathematical Expression	Units & Applicability Notes
		and Tc. Temperatures MUST be in Kelvin (K = °C + 273.15).
Mayer's Specific Heat Relation	`Cp - Cv = R (and Cp,m - Cv,m = R / M)`	R = 8.314 J/mol·K (universal gas constant). Difference represents external work done during isobaric expansion.

Topper Detailed Proofs & Deep Dive

Rigorous analytical study notes for Class 12 Physics: Thermodynamics, covering the First Law, energy accounting, thermodynamic processes, calculus derivations, heat engine mechanics, Second Law statements, and the four-stage Carnot cycle.

1. Foundations of Thermodynamics, State Variables & The First Law [00:00 - 12:30]

Thermodynamics investigates the macroscopic transformations of heat, work, and internal energy in physical systems. A thermodynamic system is characterized by macroscopic coordinates: Pressure P, Volume V, and absolute Temperature T, linked for an ideal gas by the equation of state PV = nRT. Internal energy U represents the sum of microscopic kinetic and potential energies of all molecules. In an ideal gas with zero intermolecular potential energy, Joule's law dictates that internal energy depends exclusively on temperature: U = U(T). When a gas expands by differential volume dV against an external pressure P, the mechanical work done by the system is dW = P dV. The First Law of Thermodynamics formulates the conservation of energy: the infinitesimal heat dQ absorbed by a system is apportioned between increasing its internal energy dU and performing mechanical work dW on its surroundings: dQ = dU + dW = dU + P dV. Under the standard physics sign convention: work done BY the gas (expansion) is positive (W > 0), work done ON the gas (compression) is negative (W < 0), heat absorbed is positive (Q > 0), and heat released is negative (Q < 0). Because U is an exact differential (state function), its By contrast, heat Q and work W are path-dependent quantities represented by inexact differentials.

> *Takeaway: In every closed thermodynamic cycle, net change in internal energy is ZERO (Delta U_cycle = 0), meaning net heat absorbed equals net mechanical work performed: Q_net = W_net.*

2. The Four Fundamental Thermodynamic Processes & P-V Indicator Analysis [12:30 - 19:15]

A thermodynamic process represents the progression of a system from an initial equilibrium state to a final equilibrium state along a defined thermodynamic path. Four classical quasi-static processes govern ideal gas behavior: (1) Isothermal Process (T = constant, Delta T = 0): Because internal energy depends solely on temperature, Delta U = 0, which implies Q = W. The process follows Boyle's law (P V = constant). On a P-V diagram, it forms an equilateral hyperbola with slope (dP/dV)_iso = -P/V. (2) Adiabatic Process (Q = 0): The system is thermally insulated from its surroundings (or the expansion/compression occurs so rapidly that heat exchange is negligible). From the First Law, W = -Delta U: work done during adiabatic expansion is performed entirely at the expense of internal energy, causing a substantial temperature drop. The process is governed by Poisson's relations: P V^gamma = const, T V^(gamma-1) = const, and T^gamma P^(1-gamma) = const, where gamma = Cp/Cv > 1. The slope on a P-V diagram is (dP/dV)_adi = -gamma (P/V) = gamma x (dP/dV)_iso, proving that adiabatic curves are always steeper than isothermal curves through the same state point. (3) Isobaric Process (P = constant): The gas expands or contracts under fixed external pressure. Work done is W = P(V2 - V1) = nR(T2 - T1). Heat supplied is Q = n Cp Delta T. (4) Isochoric Process (V = constant, dV = 0): Because boundaries do not displace, boundary work is identically zero: W = 0. All absorbed heat increases the internal energy of the gas: Q = Delta U = n Cv Delta T.

> *Takeaway: The slope of an adiabatic curve on a P-V diagram is gamma times steeper than an isothermal curve intersecting at the same point. Adiabatic expansion always cools the gas, whereas adiabatic compression heats it.*

3. Heat Engines, Energy Flow & The Second Law of Thermodynamics [19:15 - 26:00]

A heat engine is a cyclic device that converts thermal energy into continuous mechanical work. It operates between two thermal reservoirs: a heat Source maintained at a constant high temperature Th and a heat Sink maintained at a constant low temperature Tc (Th > Tc). In each operational cycle: (a) the working substance absorbs thermal energy Qh from the hot source; (b) the working substance expands, performing external mechanical work W; (c) the working substance rejects residual waste heat Qc into the cold sink; (d) the working substance is returned to its initial thermodynamic coordinates (P1, V1, T1). Because the engine returns to its original state, net internal energy change across one complete cycle is zero: Delta U_cycle = 0. Energy conservation gives the net mechanical work delivered: W = Qh - Qc. The thermal efficiency eta is defined as the fraction of absorbed heat converted into useful work: eta = W / Qh = (Qh - Qc) / Qh = 1 - (Qc / Qh). The Second Law of Thermodynamics dictates that thermal efficiency can never equal 100%: Kelvin-Planck Statement: It is physically impossible to construct an engine operating in a cycle whose sole effect is to extract heat from a single thermal reservoir and convert it completely into work. Clausius Statement: It is impossible to construct a cyclic machine whose sole effect is the transfer of heat from a colder body to a hotter body without an external work input.

Takeaway: Every real and ideal heat engine must reject a finite quantity of heat Qc > 0 to a low-temperature sink. A 100% efficient heat engine would violate the Kelvin-Planck statement of the Second Law.

4. The Carnot Cycle, Four Reversible Stages & Carnot's Theorem [26:00 - 38:00]

In 1824, Nicolas Léonard Sadi Carnot demonstrated that maximum thermal efficiency is achieved by an idealized, reversible cycle operating between two temperature limits Th and Tc. The Carnot Cycle consists of four successive reversible operations plotted on an indicator (P-V) diagram: Stage 1 (A -> B): Reversible Isothermal Expansion at Source temperature Th. The cylinder is placed on the hot source. The gas absorbs heat Qh and expands from VA to VB doing work W1 = nR Th ln(VB / VA). Stage 2 (B -> C): Reversible Adiabatic Expansion from Th to Tc. The cylinder is placed on a non-conducting stand. Gas expands thermally insulated from VB to VC, doing work W2 = nR (Th - Tc) / (gamma - 1) while temperature falls from Th to Tc. Stage 3 (C -> D): Reversible Isothermal Compression at Sink temperature Tc. The cylinder is placed on the cold sink. The gas is compressed from VC to VD, rejecting heat Qc = nR Tc ln(VC / VD) and doing work W3 = -nR Tc ln(VC / VD). Stage 4 (D -> A): Reversible Adiabatic Compression from Tc to Th. The cylinder is placed on the insulating stand. The gas is compressed from VD to VA, absorbing work W4 = -nR (Th - Tc) / (gamma - 1) and restoring the gas to state A (Th, VA). Net mechanical work per cycle is W_net = W1 + W2 + W3 + W4 = W1 + W3. Because processes B -> C and D -> A are adiabatic: Th VB^(gamma-1) = Tc VC^(gamma-1) and Th VA^(gamma-1) = Tc VD^(gamma-1). Dividing these two equations gives (VB / VA)^(gamma-1) = (VC / VD)^(gamma-1). Substituting this ratio reveals: Qc / Qh = Tc / Th! Thus, the Carnot efficiency depends solely on reservoir absolute temperatures: eta_carnot = 1 - (Tc / Th). Carnot Theorem: (1) No engine operating between two given temperatures can be more efficient than a reversible Carnot engine operating between the same two temperatures. (2) All reversible engines operating between the same two temperatures have identical efficiency, completely independent of the working substance.

Takeaway: Carnot efficiency eta = 1 - (Tc / Th) represents the theoretical ceiling of thermal efficiency. It can reach 1.0 (100%) only if the sink is at absolute zero (Tc = 0 K), which is forbidden by the Third Law of Thermodynamics.

Mathematical Derivation: Derivation 1: Work Done in Reversible Isothermal Expansion of an Ideal Gas

-> 1. Identify governing boundary conditions: Temperature T is held constant (isothermal), meaning Delta T = 0 and Delta U = 0 for an ideal gas.

-> 2. Write differential mechanical work: dW = P dV, where P is the internal pressure of the gas during quasi-static expansion.

-> 3. Substitute the ideal gas equation of state: P = (n R T) / V.

-> 4. Set up definite integral from initial volume V1 to final volume V2: W = Integral [V1 to V2] P dV = Integral [V1 to V2] (n R T / V) dV.

-> 5. Factor out constant quantities (n, R, T): W = n R T Integral [V1 to V2] (1 / V) dV.

-> 6. Evaluate the integral using standard calculus anti-derivative Integral (1/V) dV = ln(V): W = n R T [ln(V)]_[V1 to V2] = n R T [ln(V2) - ln(V1)].

-> 7. Apply logarithmic quotient identity ln(a) - ln(b) = ln(a / b): W = n R T ln(V2 / V1).

-> 8. Convert natural logarithm (base e) to common logarithm (base 10) by multiplying by ln(10) ~= 2.302585: W = 2.303 n R T log10(V2 / V1).

-> 9. Using Boyle's law (P1 V1 = P2 V2 !Ò V2 / V1 = P1 / P2) n R T log10(P1 / P2).

[Final Equation]: W = n R T ln(V2 / V1) = 2.303 n R T log10(V2 / V1) = 2.303 n R T log10(P1 / P2)

Mathematical Derivation: Derivation 2: Work Done in Reversible Adiabatic Expansion of an Ideal Gas

-> 1. Identify adiabatic condition: Q = 0. By First Law, dQ

-> 2. Express differential work: dW = -n Cv dT. Integrating between T1 and T2: W = -n Cv (T2 - T1) = n Cv (T1 - T2).

-> 3. Using Mayer's relation Cp - Cv = R and definition of adiabatic ratio gamma = Cp / Cv: Cv = R / (gamma - 1).

-> 4. Substitute Cv: W = n [R / (gamma - 1)] (T1 - T2) = [n R (T1 - T2)] / (gamma - 1).

-> 5. Alternatively, use Poisson's relation P V^gamma = K (

-> 6. Set up integral: W = Integral [V1 to V2] K V^(-gamma) dV = K [V^(1-gamma) / (1-gamma)]_[V1 to V2] = [K V2^(1-gamma) - K V1^(1-gamma)] / (1-gamma).

-> 7. Since K = P1 V1^gamma = P2 V2^gamma: K V2^(1-gamma) = P2 V2 and K V1^(1-gamma) = P1 V1.

-> 8. Therefore: W = (P2 V2 - P1 V1) / (1-gamma) = (P1 V1 - P2 V2) / (gamma - 1).

[Final Equation]: W = (P1 V1 - P2 V2) / (gamma - 1) = [n R (T1 - T2)] / (gamma - 1)

Mathematical Derivation: Derivation 3: Thermal Efficiency of the Ideal Carnot Engine Cycle

-> 1. Stage 1 (Isothermal expansion at Th from VA to VB): Heat absorbed Qh = W1 = n R Th ln(VB / VA).

-> 2. Stage 2 (Adiabatic expansion from Th to Tc, VB to VC): Work W2 = n R (Th - Tc) / (gamma - 1). Adiabatic relation: Th VB^(gamma-1) = Tc VC^(gamma-1) ... (Eq. 1).

-> 3. Stage 3 (Isothermal compression at Tc from VC to VD): Heat rejected Qc = -W3 = n R Tc ln(VC / VD).

-> 4. Stage 4 (Adiabatic compression from Tc to Th, VD to VA): Work W4 = -n R (Th - Tc) / (gamma - 1). Adiabatic relation: Tc VD^(gamma-1) = Th VA^(gamma-1) ... (Eq. 2).

-> 5. Net work per cycle: W_net = W1 + W2 + W3 + W4. Note that W2 + W4 = 0, so W_net = W1 + W3 = Qh - Qc.

-> 6. Divide Eq. 1 by Eq. 2: [Th VB^(gamma-1)] / [Th VA^(gamma-1)] = [Tc VC^(gamma-1)] / [Tc VD^(gamma-1)] !Ò (V B / V A) ^ (g a m m a - 1) = (V C / V D) ^ (g a m m a - 1) .

-> 7. Taking the (1 / (gamma-1))-th power of both sides yields: VB / VA = VC / VD.

-> 8. Compute ratio of rejected heat to absorbed heat: Qc / Qh = [n R Tc ln(VC / VD)] / [n R Th ln(VB / VA)]. Since VB/VA = VC/VD, logarithms cancel: Qc / Qh = Tc / Th.

-> 9. Substitute into thermal efficiency definition eta = 1 - (Qc / Qh): eta_carnot = 1 - (Tc / Th).

[Final Equation]: eta_carnot = 1 - (Qc / Qh) = 1 - (Tc / Th)

Examiner Scoring Pro-Tips & Pitfalls:

* Temperature Units Trap: When using eta = 1 - (Tc / Th), temperatures MUST always be converted to absolute Kelvin (K = °C + 273.15). Using Celsius will yield completely incorrect answers.

* Cyclic Invariance: Never calculate Delta U across a full cycle — for any cyclic operation, Delta U_cycle is strictly 0 J because internal energy is a state function.

* P-V Area Interpretation: The enclosed area on an indicator diagram equals net work. Clockwise cycles represent heat engines (W_net > 0); counter-clockwise cycles represent refrigerators/heat pumps (W_net < 0).

* Isothermal vs Adiabatic Slope: If asked to identify curves on a P-V graph, remember that the steeper curve is always adiabatic because its slope is gamma times that of the isothermal curve.

Chatpata Hinglish Analogies & Memory Tricks

Bhai dekho, Thermodynamics koi rocket science nahi hai! Bas simple zindgi ke hisaab-kitab se samjho: jitni heat energy doge, thoda hissa internal energy bank balance mein jama hoga aur baki se bahar doston ke saath mechanical kaam hoga.

Concept: First Law of Thermodynamics (Q = Delta U + W)

Mummy ne Pocket Money di Rs. 1000 (Heat Added, Q = +1000 J). Tumne Rs. 400 chupke se apni Gullak mein bacha liye (Internal Energy increase, Delta U = +400 J), aur baki Rs. 600 se canteen mein doston ko samose-chai ki party de di (Work Done by system, W = +600 J). Equation 100% balanced: Mummy ka paisa = Gullak + Party!

[Tip] Punchline: Gullak (Delta U) + Canteen Party (W) = Mummy Ki Pocket Money (Q)!

Concept: Isothermal vs Adiabatic Expansion

Cycle ke tyre mein aaram-aaram se hawa bharo toh garmi bahar nikal jaati hai aur temperature constant rehta hai (Isothermal). Lekin agar cycle ka tube achanak "THAAN!" karke phat jaye, toh gas ko heat lene ka time hi nahi milta (Q = 0). Gas apni hi internal energy kharch karke expand hoti hai, isliye tube ke valve par chhu kar dekho toh bilkul baraf jaisi thandi lagti hai (Adiabatic cooling)!

[Tip] Punchline: Aaram se = Isothermal; Achanak Blast = Adiabatic Thandak!

Concept: Isochoric Process (Volume Constant, W = 0)

Ghar ka pressure cooker imagine karo jiski seeti jam ho gayi hai. Steel ki body rigid hai, toh volume hil hi nahi sakta (dV = 0). Jab volume hil hi nahi sakta toh piston ka koi work nahi hoga (Work Done = 0). Jitni gas jalaoge, saari energy andar ke daal-chawal aur steam ka pressure badhane mein chali jayegi.

[Tip] Punchline: Iso-CHORIC matlab Volume bilkul CHUP (Work = 0)!

Concept: Heat Engine & Why 100% Efficiency Is Impossible

Yeh bilkul school ke Sharma ji ke bete jaisa hai! Chahe kitna bhi padh le, 100% full attendance aur zero mistake exam mein impossible hai. Second Law kehta hai: engine kitna bhi mehanga bana lo, thoda na thoda dhuan aur heat sink mein waste karni hi padegi. Zero waste engine banana kudrat ke khilaaf hai!

[Tip] Punchline: 100% efficiency is a myth, bro! Thoda loss toh Sharma ji ke bete ka bhi hota hai!

Concept: Carnot Engine (The Ideal Topper)

Carnot Engine physics ki duniya ka "Ideal Topper" hai jo paper par exist karta hai lekin real life mein dhundhne se bhi nahi milta! Chaar step mein ghumta hai: pehle source se heat kheecho (isothermal expansion), phir akele maze se expand karo (adiabatic expansion), phir kachra thande sink mein phenko (isothermal compression), aur wapas apni purani jagah par aa jao (adiabatic compression).

[Tip] Punchline: Carnot engine = Theoretical Topper. Real life mein friction aur heat loss zaroor aayenge!

Key Desi Takeaways (Bhai Ka Summary)

-> Indicator diagram ka andar ka area = Total mechanical work done. Agar chakkar clockwise hai toh engine kaam kar raha hai (fayda); anti-clockwise hai toh refrigerator hai (bahar se bijli kharch hogi).

-> Kelvin Scale Rule: Formula mein kabhi bhi degree Celsius mat daalna bhai, warna marks ka kalyan ho jayega! Always Kelvin = °C + 273.15.

-> Cyclic Process Magic: Full cycle ke baad Delta U hamesha ZERO hota hai. Sawal mein cyclic dekho toh seedha Delta U = 0 likho!

Fast Exam Recognition (Jugaad Points):

* Paper mein agar likha ho "sudden burst" ya "fast process", toh 100% Adiabatic equation (P V^gamma = const) lagana!

* Agar likha ho "slowly conducting cylinder", toh aankh band karke Isothermal (T = const) lagana!

* Efficiency formula yaad rakhne ka tareeka: 1 minus (Chhota Temp / Bada Temp) in Kelvin!

Practice Questions & Evaluation Rubrics

1 MARK *AI Practice Question — Board Format (1 Mark)*

Q: State the First Law of Thermodynamics and write its mathematical equation in differential form, defining all symbols.

Answer:
The First Law of Thermodynamics states that energy can neither be created nor destroyed; heat supplied to a closed system equals the sum of the increase in its internal energy and the work done by the system on its surroundings.
Mathematical formulation: dQ = dU + dW = dU + P dV
Where: dQ = infinitesimal heat absorbed by the system (Joules), dU = infinitesimal change in internal energy (Joules), and dW = P dV = mechanical work done by the system against external pressure P (Joules).

Evaluation Rubric:
- Correct verbal statement of First Law (½ mark)
- Correct differential equation with explicit symbol definitions (½ mark)

3 MARK *AI Practice Question — Board Format (3 Marks)*

Q: Derive an analytical expression for the work done during the reversible isothermal expansion of n moles of an ideal gas from volume V1 to volume V2 at temperature T.

Answer:
1. Condition: In an isothermal process, temperature T remains constant (Delta T = 0), so Delta U = 0 for an ideal gas.
2. Work done for differential volume change dV: dW = P dV.
3. Using the ideal gas equation of state: P = n R T / V.
4. Integrating from initial volume V1 to final volume V2:
W = Integral [V1 to V2] P dV = Integral [V1 to V2] (n R T / V) dV = n R T Integral [V1 to V2] (1 / V) dV
5. Evaluating the integral: W = n R T [ln(V)]_[V1 to V2] = n R T ln(V2 / V1).
6. Converting to base-10 logarithm:
W = 2.303 n R T log10(V2 / V1).
7. In terms of pressures (using Boyle's law P1 V1 = P2 V2 !Ò V2 / V1 = P1 / P2):
W = 2.303 n R T log10(P1 / P2).

Evaluation Rubric:
- Setting up the integral dW = P dV with ideal gas substitution P = nRT/V (1 mark)
- Correct integration limits and natural logarithm step nRT ln(V2/V1) (1 mark)
- Final formula with 2.303 conversion factor and pressure equivalence (1 mark)

5 MARK *AI Practice Question — Board Format (5 Marks)*

Q: Describe the working of a Carnot engine. Explain the four reversible stages of the Carnot cycle with the aid of a neat P-V indicator diagram, and derive the expression for its thermal efficiency eta = 1 - (Tc / Th).

Answer:
A Carnot engine is an idealized, reversible heat engine operating between a heat source at Th and a heat sink at Tc.
The four successive stages on the P-V diagram are:
1. Stage 1 (A -> B): Reversible Isothermal Expansion at constant source temperature Th. Gas absorbs heat Qh from the source and expands from VA to VB. Work done: W1 = Qh = n R Th ln(VB / VA).
2. Stage 2 (B -> C): Reversible Adiabatic Expansion from Th to Tc. Gas expands without heat exchange from VB to VC, doing work W2 = n R (Th - Tc) / (gamma - 1). Adiabatic relation: Th VB^(gamma-1) = Tc VC^(gamma-1) ... (Eq. 1).
3. Stage 3 (C -> D): Reversible Isothermal Compression at constant sink temperature Tc. Gas is compressed from VC to VD, rejecting heat Qc to the sink. Work done: W3 = -Qc = -n R Tc ln(VC / VD).
4. Stage 4 (D -> A): Reversible Adiabatic Compression from Tc to Th. Gas is compressed without heat exchange from VD to VA, restoring initial state A. Work done: W4 = -n R (Th - Tc) / (gamma - 1). Adiabatic relation: Tc VD^(gamma-1) = Th VA^(gamma-1) ... (Eq. 2).

Net Work & Efficiency Derivation:
Net work: W_net = W1 + W2 + W3 + W4. Since W2 + W4 = 0, W_net = W1 + W3 = Qh - Qc.
Dividing Eq. 1 by Eq. 2: (VB / VA)^(gamma-1) = (VC / VD)^(gamma-1) !Ò VB / VA = VC / VD.
Therefore, the heat ratio simplifies to: Qc / Qh = [n R Tc ln(VC / VD)] / [n R Th ln(VB / VA)] = Tc / Th.
Thermal efficiency is defined as: eta = W_net / Qh = (Qh - Qc) / Qh = 1 - (Qc / Qh).
Substituting Qc / Qh = Tc / Th:
eta_carnot = 1 - (Tc / Th).

Diagnostic Assessment & Answer Key

1. In any closed cyclic thermodynamic process, what is the net change in internal energy (Delta U_cycle)?

(A) Zero (Delta U = 0)
(B) Equal to the work done (Delta U = W)
(C) Equal to total heat absorbed (Delta U = Qh)
(D) Depends on the working substance

Answer: (A) • Topic: First Law & State Functions
Reason: Because internal energy U is a state function that depends only on equilibrium temperature, a system returning to its initial state has Delta U_cycle = 0.

2. During reversible adiabatic expansion of an ideal gas, work is performed by the gas at the expense of what?

(A) Its own internal energy (causing temperature to decrease)
(B) External heat supplied from the surroundings
(C) Gravitational potential energy
(D) Atmospheric friction

Answer: (A) • Topic: Adiabatic Process
Reason: In an adiabatic process Q = 0, so by the First Law W = -Delta U. Work done by the gas reduces its internal energy, resulting in a temperature drop.

3. A Carnot engine operates between a source at 500 K and a sink at 300 K. What is its thermal efficiency?

(A) 40% (0.40)
(B) 60% (0.60)
(C) 20% (0.20)
(D) 100% (1.00)

Answer: (A) • Topic: Carnot Cycle Efficiency
Reason: Efficiency eta = 1 - (Tc / Th) = 1 - (300 / 500) = 1 - 0.60 = 0.40, which equals 40%.

4. Which statement correctly compares the slopes of adiabatic and isothermal curves passing through the same state on a P-V diagram?

(A) The adiabatic curve is steeper by a factor of gamma (Cp/Cv)
(B) The isothermal curve is steeper by a factor of gamma
(C) Both curves have identical slopes
(D) The adiabatic curve is horizontal

Answer: (A) • Topic: Indicator Diagram Slopes
Reason: The slope of an isothermal curve is dP/dV = -P/V, while for an adiabatic curve dP/dV = -gamma(P/V). Because gamma > 1, the adiabatic curve is gamma times steeper.

5. Under the standard Physics sign convention, what is the sign of work done when a gas is compressed by external force?

(A) Negative (W < 0)
(B) Positive (W > 0)
(C) Zero (W = 0)
(D) Infinite

Answer: (A) • Topic: Sign Conventions
Reason: In Physics, work done BY the gas during expansion (dV > 0) is positive; work done ON the gas during compression (dV < 0) is negative.

Lecture Roadmap & Smart Timestamps

Timestamp	Lecture Segment	Core Concept Summary
00:00	**Foundations & State Variables (P, V, T, U)**	Equilibrium states, system boundaries, and ideal gas equation of state.
05:45	**First Law of Thermodynamics & Sign Conventions**	Internal energy, conservation of energy (Q = Delta U + W), and work dW = P dV.
12:30	**Thermodynamic Processes & Isothermal Work Derivation**	Isothermal, adiabatic, isobaric, and isochoric paths with calculus derivations.
19:15	**Heat Engines & The Second Law of Thermodynamics**	Working substance, source/sink reservoirs, thermal efficiency, and Kelvin-Planck & Clausius statements.
26:00	**The Carnot Cycle: Four Reversible Operations**	P-V indicator analysis, 2 isothermals + 2 adiabatics, and net work calculation.
33:20	**Carnot Theorem, Reversibility & Numerical Application**	Maximum theoretical efficiency (eta = 1 - Tc/Th) and step-by-step 500K-300K numerical.